ALICE L WALTON SCHOOL OF MEDICINE

Peritonitis

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Adapted from: Peritonitis

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Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Differentiate between primary, secondary, and peritoneal dialysis peritonitis based on etiology, microbial patterns, and clinical presentation.
2. Identify the common causative organisms responsible for spontaneous itis, and dialysis-asso
3. Describe the presentation of peritonitis and the diagnostic approach.
4. Outline the treatment strategies for each type of peritonitis, including appropriate antibiotic regimens and the indications for surgical source control.
5. Differentiate the etiologies and clinical presentation of carcinomatosis, peritoneal mesothelioma, and pseudomyxoma peritonei.

CASE CONNECTION

A 72-year-old male with a history of cirrhosis complicated by ascites presents to the ED with abdominal pain. He is febrile and tachycardic. He is lying still on the ED stretcher. His abdomen is distended and has diffuse tenderness to percussion. Palpation reveals involuntary guarding.

Paracentesis is performed for cell count and culture of his ascites fluid. The fluid is cloudy with a cell count of 1,200 WBC/mm^3.

Cefotaxime is started for Gram-negative coverage and the patient is admitted. The following day, he has worsening fever, tachycardia, and is now hypotensive. The microbiology lab is called for culture results. Their preliminary results show growth of multiple bacterial species.

What do you do next?

GO TO CONCLUSION ↓

What Is Peritonitis? (LO 1)

Peritoneum, the lining of the abdominal cavity, is made of simple squamous epithelium (mesothelium) supported by a thin layer of connective tissue. The visceral peritoneum is adherent to the intraperitoneal organs and the parietal peritoneum lines the abdominal wall (Figure 1). The space between these two layers is called the peritoneal cavity and contains a small amount of slippery serous fluid that allows these two layers to slide freely over each other.

Peritonitis is inflammation of the peritoneum, typically due to infection from bacteria. This can be classified as primary or secondary.

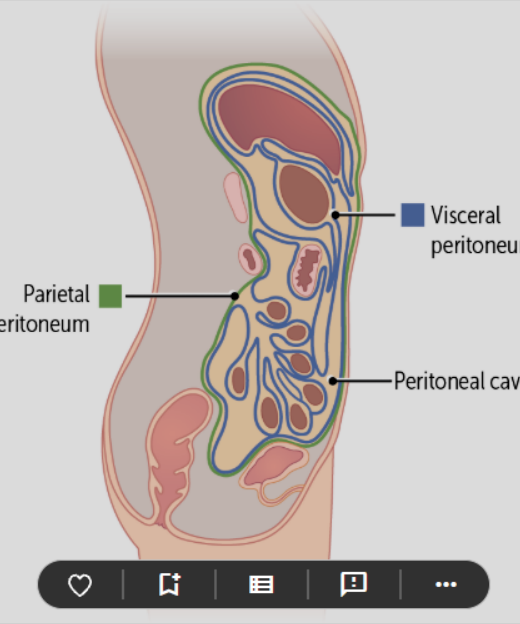


Figure 1

Types of Peritonitis (LO 1, 2)

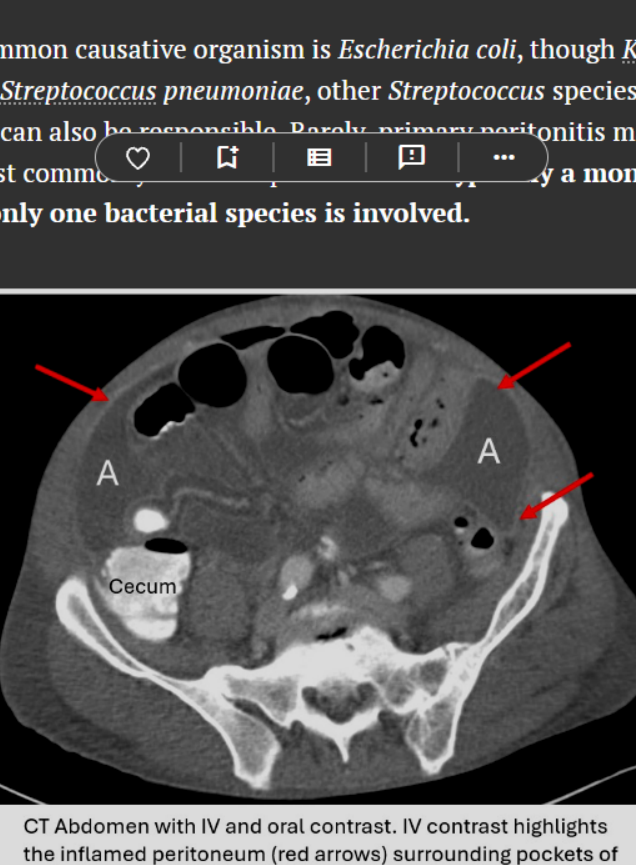
Primary Peritonitis/Spontaneous Bacterial Peritonitis

Primary peritonitis is infection of the peritoneal cavity without an identifiable intra-abdominal source and is commonly referred to as **spontaneous bacterial peritonitis (SBP)**. It most often occurs as a **complication of ascites** in patients with cirrhosis (Figure 2). The usual mechanism is bacterial translocation across the intestinal wall. Less commonly, infection occurs via hematogenous spread.

CLINICAL CORRELATION

Bacterial translocation can occur in healthy individuals but is normally controlled by peritoneal defenses, including macrophages, complement, and neutrophils. In ascites, these defenses are diluted, allowing bacteria to proliferate and spread within the peritoneal cavity.

The most common causative organism is *Escherichia coli*, though *Klebsiella pneumoniae*, *Streptococcus pneumoniae*, other *Streptococcus* species, and *Enterobacter* can also be responsible. Rarely, primary peritonitis may be caused by fungi, most commo **y a monomicrobial infection - only one bacterial species is involved.**



CT Abdomen with IV and oral contrast. IV contrast highlights the inflamed peritoneum (red arrows) surrounding pockets of infected ascites (A) due to increased blood flow.

Figure 2. CT Abdomen of a patient with SBP (CT scans are not typically needed for this diagnosis.)

What is the most c involved in primary peritonitis?

Secondary Peritonitis

Secondary peritonitis is defined as inflammation and infection of the peritoneal cavity resulting from a **disruption of gastrointestinal (or genitourinary) tract integrity**, allowing direct contamination of the peritoneal space by endogenous microflora (Figure 3). It most commonly involves perforation of a hollow viscus due to disease (e.g., perforated appendicitis, diverticulitis, peptic ulcer), trauma, anastomotic dehiscence, or ischemic bowel injury.

The infection is characteristically **polymicrobial**, involving a **mixture of aerobic and anaerobic organisms reflecting the flora of the disrupted organ**, with common pathogens including *Escherichia coli*, *Klebsiella* spp., *Bacteroides* spp., *Enterococcus* spp., and *Clostridium* spp. Secondary peritonitis has a high mortality in those who develop severe sepsis (35%). Management requires a combination of resuscitation, source control (typically surgical exploration of the abdomen), and empiric broad-spectrum antibiotics — distinguishing it from primary peritonitis, which is generally managed with antibiotics alone.

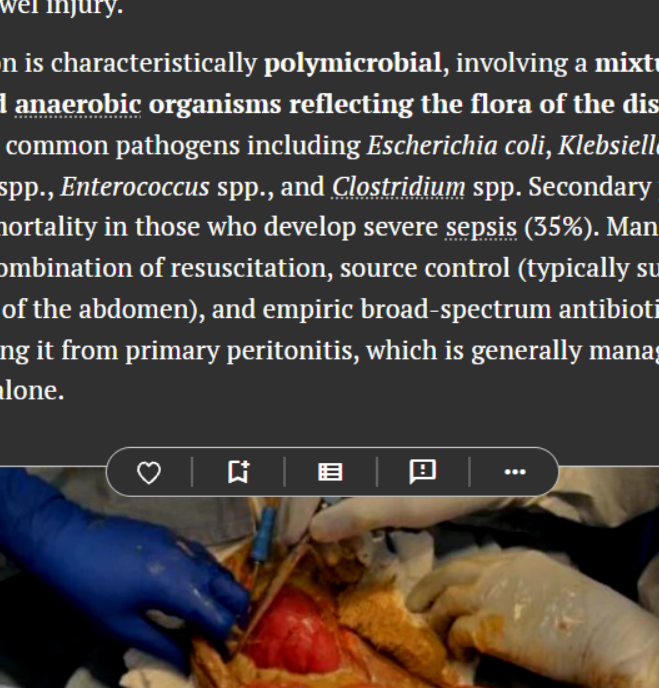


Figure 3. Peritonitis secondary to colonic perforation

CLINICAL REASONING

Source control is a universal principle in infectious disease management. It is required when infection is sustained by a localized collection of infected material or an ongoing source of contamination that antibiotics alone cannot eliminate. This may involve draining infected fluid or pus, irrigating contaminated spaces, and repairing or removing the underlying source of infection. Chest tube drainage of an empyema is another example.

Peritoneal Dialysis Associated Peritonitis

To perform peritoneal dialysis, a catheter is inserted into the peritoneal cavity. Dialysis fluid (dialysate) is infused and allowed to dwell, where concentration gradients drive diffusion of metabolic waste products across the peritoneum into the fluid (Figure 4). The dialysate is then drained and replaced, and the cycle is repeated multiple times each night.

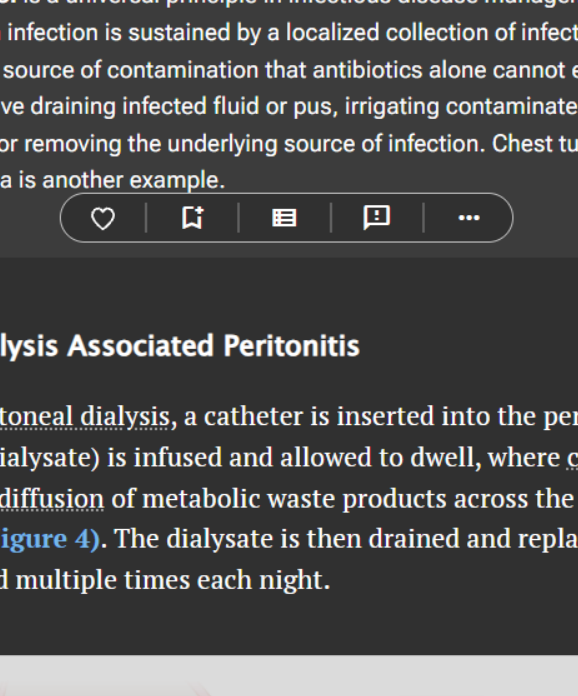


Figure 4. Peritoneal Dialysis

Peritoneal dialysis–associated peritonitis does not meet criteria for primary peritonitis due to external access via the catheter, nor for classic secondary peritonitis, as there is typically no perforated viscus or surgically correctable source. It is therefore considered a distinct category. **Infection most commonly occurs when bacteria (usually skin flora) or fungi enter the peritoneal cavity through or around the catheter.** Patients present with abdominal pain and signs of peritonitis. The dialysate is often cloudy due to inflammatory cells. A sample should be sent for culture and sensitivities.

Bacterial dialysis-associated peritonitis, like primary peritonitis, is usually monomicrobial. The most common causes are *Staphylococcus aureus* or *Staphylococcus epidermidis* from skin accessing through the catheter, yet Gram-negative rods cause a substantial number of cases due to intestinal translocation (similar to SBP). The most common fungal cause is *Candida albicans*.

Clinical Presentation of Peritonitis (LO 3)

Symptoms

Early in the disease process, visceral pain is transmitted by slow, unmyelinated C fibers on the walls of hollow viscera. **Visceral pain is a dull, poorly localized, midline pain.** The parietal peritoneum is somatic - part of the abdominal wall. It is richly innervated with fast, myelinated A-delta fibers. **Somatic pain is sharp, well-localized, and exacerbated by any maneuver that stretches the peritoneum.** Many disorders, such as appendicitis, start with poorly localized to a more focal area as the inflammation his transition from visceral to somatic pain is the physiologic basis for localized "peritoneal signs."

CLINICAL REASONING

Visceral pain is frequently accompanied by **poor appetite, nausea, and sometimes vomiting** because of transmission of sensory signals to the brainstem centers. These symptoms remain when the pain progresses to somatic, because the viscera are still inflamed.

Signs

The physical exam of peritonitis may reveal fever and possibly hypotension if sepsis progresses to septic shock. The patient will typically lay still because any form of movement will disturb the peritoneum and worsen the pain. A patient writhing in pain is unlikely to have peritonitis (a symptomatic kidney stone is more likely).

Findings on abdominal examination may reveal focal peritonitis, as in uncomplicated appendicitis, or diffuse peritonitis, as may occur with gross spillage of intestinal contents. The components in the abdominal examination in peritonitis are as follows:

- **Inspection:** The abdomen may be distended in the setting of ascites, ileus, or bowel obstruction.
- **Auscultation:** For standardized examinations, you will likely see reports of "hypoactive" o e it as a clue for peritonitis. In the real world, auscultation of bowel sounds for peritonitis has tremendous interobserver variability and zero utility. In the setting of bowel obstruction, however, high-pitched, tinkling bowel sounds may be heard.
- **Percussion:** Focal percussion tenderness is a sensitive and gentle maneuver to elicit tenderness from peritonitis, and to determine the point of maximal tenderness. Tympany on percussion suggests air-filled intestine.
- **Palpation:** Begin the palpation away from the location of reported pain. Assess for relaxation of the abdominal wall muscles. Voluntary guarding implies the patient is consciously contracting the abdominal wall muscles to minimize deep palpation and avoid worsened pain. It can often be overcome by distraction. **Involuntary guarding, synonymous with rigidity, implies peritonitis - it is the most specific examination finding for peritonitis.** Parietal peritoneal inflammation initiates a spinal reflex arc, leading to involuntary abdominal muscle contraction in an attempt to limit peritoneal irritation. The resulting rigidity of the abdominal muscles cannot be relaxed.

CLINICAL REASONING

Rebound tenderness: Elicited by pressing deeply enough to depress the peritoneum, holding pressure for several seconds, then releasing suddenly — pain is worse on **release** rather than compression. This maneuver directly stretches the inflamed peritoneum and causes *exquisite* pain.

Although classically described, it is rarely elicited in practice due to patient discomfort.

You may find descriptions of rebound tenderness on standardized exams and should equate the finding with peritonitis. On real patients, focus instead on evaluating for percussion tenderness and rigidity (involuntary guarding).

The **physical exam findings in spontaneous bacterial peritonitis are less severe.** Because of the lower bacterial burden, the lack of intestinal contents (bile, gastric acid, fecal material), and the physical buffer of ascitic fluid, SBP may present with generalized tenderness and percussion tenderness but is unlikely to produce the involuntary rigidity characteristic of secondary peritonitis. In fact, rigidity in a patient with ascites is concerning for secondary peritonitis.

What are some physical examination signs of peritonitis?

How Is Peritonitis Diagnosed? (LO 3)

Patients with spontaneous bacterial peritonitis and peritoneal dialysis-associated peritonitis are diagnosed clinically based on signs and symptoms of peritonitis in the appropriate clinical setting. Peritoneal fluid is sampled in both conditions for cell count and culture, by paracentesis of ascites fluid, or draining fluid from the peritoneal dialysis catheter.

The diagnosis of secondary peritonitis is also primarily clinical. However, confidently identifying patients who need emergent surgical treatment based on physical examination takes years of experience. Over the last 25 years, abdominal imaging has become increasingly common and is now ubiquitous. CT imaging in particular has made identification of the underlying cause of peritonitis more straightforward.

Plain films

Plain abdominal radiographs have a limited but still relevant role in the evaluation of peritonitis. They may reveal findings suggestive of an underlying cause, such as bowel obstruction or perforation and resultant free intraabdominal air. However, plain films are not sufficient to exclude gastrointestinal perforation, and CT imaging is often required when clinical suspicion remains high.

CLINICAL CORRELATION

An **upright chest radiograph** is particularly useful for detecting free intraperitoneal air u **5% sensitive** for free air from a perforated duodenal ulcer (Figure 5).

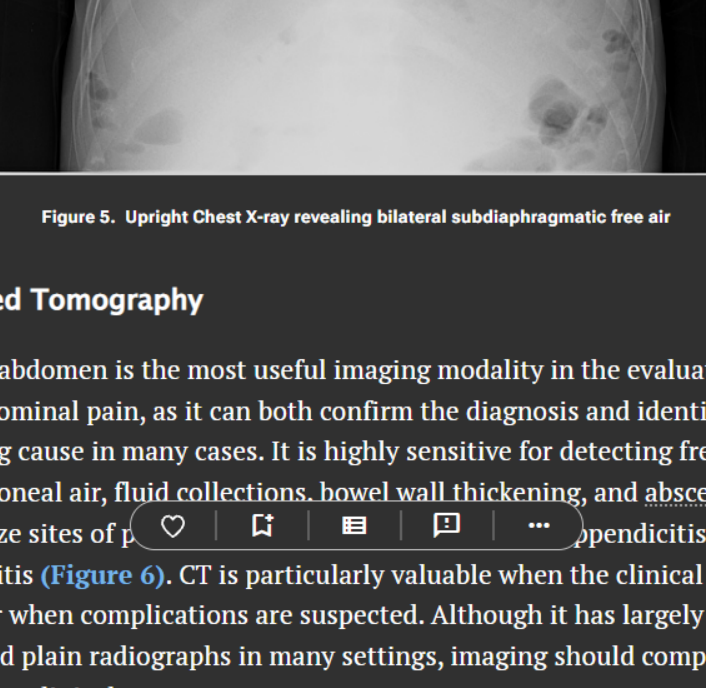


Figure 5. Upright Chest X-ray revealing bilateral subdiaphragmatic free air

Computed Tomography

CT of the abdomen is the most useful imaging modality in the evaluation of acute abdominal pain, as it can both confirm the diagnosis and identify the underlying cause in many cases. It is highly sensitive for detecting free intraperitoneal air, fluid collections, bowel wall thickening, and abscesses, and can localize sites of p ppendicitis or diverticulitis (Figure 6). CT is particularly valuable when the clinical picture is unclear or when complications are suspected. Although it has largely supplanted plain radiographs in many settings, imaging should complement, not replace, clinical assessment.

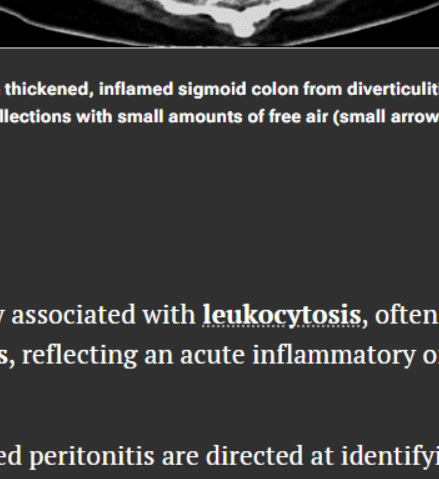


Figure 6. CT Abdomen with thickened, inflamed sigmoid colon from diverticulitis (large arrows). Free fluid collections with small amounts of free air (small arrows)

Labs

Peritonitis is typically associated with **leukocytosis**, often with a **left shift** and increased **band forms**, reflecting an acute inflammatory or infectious response.

Other labs in suspected peritonitis are directed at identifying the underlying cause, including amylase and lipase for pancreatitis and liver function tests for biliary disease.

Fluid Analysis

In suspected spontaneous bacterial peritonitis (SBP), prompt diagnostic paracentesis is required. Ascitic fluid **PMN count $\geq$250 cells/mm^3** is diagnostic, even with negative cultures; fluid should also be cultured.
In peritoneal dialysis–associated peritonitis, dialysate should be inspected for cloudiness and sent for cell count and culture as well. Both conditions are typically monomicrobial.

CLINICAL REASONING

Polymicrobial cultures suggest secondary peritonitis from a perforated viscus and require urgent surgical evaluation.

Treatment of Peritonitis (LO 4)

If SBP is suspected, aggressive treatment needs to be started immediately. For SBP, because the cause is typically a bacterial infection in the peritoneum with no other source, the treatment is broad-spectrum antibiotics. The choice should cover *E. coli*, *K. pneumonia*, and other gram-negative enteric pathogens (common pathogens for SBP). Antibiotics of choice include 3rd generation cephalosporins: **Cefotaxime or Ceftriaxone**. The spectrum of coverage includes gram negative bacteria, *Haemophilus*, *E. Coli*, *Neisseria*, *Proteus*, *Klebsiella*. If the source is fungal, broad-spectrum antifungal medications (caspofungin, flucon (Table 1)

Which in HENPK are Gram-negative rods (GNRs)?

Since peritoneal dialysis peritonitis is commonly caused by Gram-positive bacteria (skin flora) and gram negative bacteria (translocation of GI bacteria), both these classes require adequate coverage. This management is more of a specialty (Nephrology) issue, so we will avoid specific antibiotic regimens. It is notable that the **antibiotics in these cases are instilled into the peritoneum via the peritoneal dialysis catheter.**

For secondary peritonitis, treatment requires both intravenous antibiotics and source control. Source control typically requires surgical intervention, such as appendectomy for a ruptured appendix, repair of a perforated intestine, or patching of a perforated duodenal ulcer. Intraabdominal irrigation is performed to reduce residual contamination. **IV antibiotics alone wi ecause there is ongoing peritoneal**

Broad-spectrum antibiotics for secondary peritonitis must cover E*nterococci*, gram-negative organisms, and anaerobes. Treatment should be initiated in the Emergency Department and continued postoperatively. Single agents such as **piperacillin/Tazobactam or ertapenem** can provide broad gram-negative coverage with anaerobic coverage. Alternatively, a combination of **cefepime** (4th gen= broad gram-negative coverage) **with metronidazole** (good Gram-negative anaerobic coverage) can also be used. (Table 1)

Table 1: Treatment for various Peritonitis

Type of Peritonitis	Pattern	Most Common Pathogens	Antibiotic Treatment
Spontaneous Bacterial Peritonitis (ascites)	Monomicrobial	*Escherichia coli* or *Klebsiella pneumoniae*	3rd Gen Cephalosporins: Cefotaxime or Ceftriaxone
Peritoneal Dialysis-associated	Monomicrobial	*Staphylococcus epidermidis* or *Staphylococcus aureus* or *Gram-negative rods*	**Intraperitoneal instillation** Do not learn the specifics.
Secondary Peritonitis (perforation of intra-abdominal organ)	Polymicrobial	Gram-negative rods: *Escherichia coli*, *Klebsiella* spp. **and** Anaerobes: *Bacteroides fragilis* **and** Gram-positive: *Enterococcus spp.*	Piperacillin/Tazobactam or Cefepime + Metronidazole Source control (surgical treatment) is mandatory

Peritoneal Malignancies (LO 5)

1. Peritoneal Carcinomatosis

Peritoneal carcinomatosis is not a primary cancer — it is the spread of cancer from another organ (most commonly ovarian, colorectal, or gastric) to the peritoneal surfaces. Cancer cells shed from the primary tumor into the peritoneal fluid, travel with normal fluid flow to predictable sites (pelvis, paracolic gutters, subdiaphragmatic spaces), and implant on the peritoneal lining. The omentum is a particularly common site of metastatic colonization.

Clinically, peritoneal carcinomatosis presents with progressive abdominal distension from malignant ascites, weight loss, and bowel obstruction. Cytology of the fluid can confirm the diagnosis. On CT, findings include peritoneal thickening, nodularity, and "omental caking."

Treatment for selected patients involves cytoreductive surgery (CRS) combined with heated intraperitoneal chemotherapy (HIPEC). Prognosis depends heavily on the primary tumor type and extent of peritoneal disease, yet carcinomatosis is not usually curable.

2. Peritoneal Mesothelioma

Peritoneal mesothelioma is a primary malignancy arising from the mesothelial cells that line the peritoneal cavity, accounting for roughly 15–20% of all mesotheliomas. Like its pleural counterpart, it is associated with **asbestos exposure**, though the association is weaker for peritoneal disease, and many patients have no identifiable exposure history. A subset of cases is considered idiopathic.

Patients typically present with abdominal pain, increasing abdominal girth, and ascites. Diagnosis requires tissue biopsy, and the immunohistochemical panel is high-yield: mesothelioma stains **positive for calretinin** and **negative for CEA**, distinguishing it from adenocarcinoma. This applies to pleural and peritoneal disease.

For resectable disease, the standard treatment is cytoreductive surgery + HIPEC. Peritoneal mesothelioma generally carries a better prognosis than pleural mesothelioma, with median survival of several years in patients who undergo complete cytoreduction.

3. Pseudomyxoma Peritonei

Pseudomyxoma peritonei (PMP) is a clinical syndrome characterized by the progressive accumulation of **mucinous (gelatinous) ascites** throughout the peritoneal cavity — sometimes called "jelly belly." It almost always originates from a ruptured **mucinous neoplasm of the appendix**. The mucin-producing tumor cells spread along peritoneal surfaces collecting preferentially in the pelvis, omentum, paracolic gutters, and along the liver capsule.

The definitive treatment is Cytoreductive Surgery + HIPEC, which can be curative in low-grade disease.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Although the patient presented with ascites and peritonitis with the expectation of SBP, preliminary culture of the ascites fluid reveals a polymicrobial infection. This patient has secondary peritonitis.

Antibiotic treatment is add coverage of Enterococcus and anaerobes to the Gram-negative coverage. CT imaging is avoided because the patient is hemodynamically unstable from peritonitis which is now expected to be from a perforated intestine. Surgical exploration is now indicated for source control, resection of perforated bowel, and irrigation of the abdominal cavity.

Summary

- Peritonitis is inflammation of the peritoneum classified as primary (spontaneous bacterial peritonitis in the setting of ascites), secondary (perforation of an intra-abdominal organ), or peritoneal dialysis-associated, each with distinct mechanisms, microbiology, and management strategies.
- Primary (spontaneous bacterial) peritonitis is most commonly caused by **monomicrobial** gram-negative enteric bacteria, particularly *Escherichia coli* and **Klebsiella pneumoniae**, entering the peritoneal cavity via bacterial translocation in patients with cirrhosis and ascites.
- Secondary peritonitis results from disruption of gastrointestinal tract integrity and is characteristically **polymicrobial,** involving Gram-negative rods, anaerobes, and Gram-positive organisms.
- Peritoneal dialysis peritonitis is its own category and is also **monomicrobial**, involving involve skin flora entering through or around the dialysis catheter or enteric bacteria translocation.
- Clinical presentation of peritonitis includes fever, diffuse abdominal pain that transitions from visceral (dull, poorly localized) to somatic (sharp, well-localized), r
- Hallmark physical exam findings of peritonitis include percussion tenderness and involuntary guarding (rigidity).
- Diagnosis of SBP is by clinical assessment supported by peritoneal fluid analysis (PMN count $\geq$250 cells/mm^3 is diagnostic of SBP) with Gram-stain and culture.
- Secondary peritonitis is also diagnosed clinically, yet abdominal imaging such as CT is useful for identifying the underlying cause if the patient is stable.
- Treatment is tailored to peritonitis type: SBP is managed with 3rd generation cephalosporins for broad Gram-negative coverage; surgical treatment is not required.
- Secondary peritonitis requires mandatory surgical source control combined with broad-spectrum antibiotics covering Gram-negative rods, anaerobes, and *Enterococci* (e.g., Piperacillin/Tazobactam or Cefepime plus Metronidazole).
- Carcinomatosis is diffuse metastatic tumor spread throughout the peritoneal cavity, often with malignant ascites.
- Peritoneal mesothelioma is a primary tumor of the peritoneum, typically (but not always) associated with asbestos exposure, that can mimic carcinomatosis.
- Pseudomyxoma peritonei is a gelatinous malignancy accumulating in the peritoneal cavity, typically from a mucinous appendiceal neoplasm.

Review Questions

1. A patient presents to the emergency room with right lower quadrant pain. He is suspected of having a perforated duodenal ulcer. Which of the following signs is

A. ○ Abdominal distention
B. ○ Abdominal rigidity
C. ○ Fever
D. ○ Hypertension
E. ○ Pacing up and down hallway
F. ○ Voluntary guarding

Explanation (requires correct answer)

2. Which of the following can lead to primary peritonitis?

A. ○ Cirrhosis
B. ○ Diverticulitis
C. ○ Pancreatitis
D. ○ Small bowel perforation
E. ○ Trauma to the abdominal wall

Explanation (requires correct answer)

3. A peritoneal fluid laboratory culture shows growth of a Gram-positive coccus, several Gram-negative rods, and anaerobic bacteria. Which is the most likely etiology?

A. ○ ascites contamination with translocated bacteria
B. ○ bacteria entering through a peritoneal dialysis catheter
C. ○ colon perforation
D. ○ primary peritonitis
E. ○ spontaneous bacterial peritonitis

Explanation (requires correct answer)

4. A patient on the Internal Medicine service is being treated for peritonitis with intravenous Cefotaxime. No surgical treatment is planned. What is his most likely diagnosis?

A. ○ appendicitis
B. ○ primary peritonitis
C. ○ peritoneal dialysis peritonitis
D. ○ secondary peritonitis

Explanation (requires correct answer)

5. A 68-year-old woman with ovarian cancer develops diffuse tumor implants throughout the peritoneal cavity with malignant ascites. Which term best describes this pattern of disease spread?

A. ○ Peritoneal mesothelioma
B. ○ Pseudomyxoma peritonei
C. ○ Carcinomatosis
D. ○ Primary peritonitis

Explanation (requires correct answer)

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ Differentiate between primary, secondary, and peritoneal dialysis peritonitis based on etiology, microbial patterns, and clinical presentation.
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03 ☐ Describe the presentation of peritonitis and the diagnostic approach.
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05 ☐ Differentiate the etiologies and clinical presentation of carcinomatosis, peritoneal mesothelioma, and pseudomyxoma peritonei.

Objective Confidence: 0/5